KKT condition example

$$\min \quad 2x_1^2 + 4x_2^2 + x_1x_2 + 8x_3^2 + 2x_2x_3$$

s.t.

$$x_1 + x_2 + 1.5x_3 - 1.2 = 0$$

$$x_1 + x_2 + x_3 \geq 1 \qquad (\text{if } x_1+x_2+x_3 \leq 1)$$

$$x_1, x_2 \geq 0, \quad x_3 \text{ free}$$

KKT conditions: (包括下面4种类型条件)

Lagrange function

$$L = 2x_1^2 + 4x_2^2 + x_1x_2 + 8x_3^2 + 2x_2x_3 - \lambda(x_1+x_2+1.5x_3-1.2) - \mu(x_1+x_2+x_3-1) - s_1x_1 - s_2x_2$$

① Lagrange derivative condition (LDC)

$$\begin{cases} \dfrac{\partial L}{\partial x_1} = 0 \Rightarrow 4x_1 + x_2 - \lambda - \mu - s_1 = 0 \\ \dfrac{\partial L}{\partial x_2} = 0 \Rightarrow 8x_2 + x_1 + 2x_3 - \lambda - \mu - s_2 = 0 \\ \dfrac{\partial L}{\partial x_3} = 0 \Rightarrow 16x_3 + 2x_2 - 1.5\lambda - \mu = 0 \end{cases}$$

② Primal feasibility condition (or original decision constraints (ODC))

$$\begin{cases} x_1 + x_2 + 1.5x_3 - 1.2 = 0 \\ x_1 + x_2 + x_3 \geq 1 \\ x_1, x_2 \geq 0 \end{cases}$$

③ Dual feasibility condition (or multiplier sign conditions (MSC))

$$\mu, s_1, s_2 \geq 0, \quad \lambda \text{ free} \qquad (\mu \leq 0)$$

④ Complementary slackness condition (CSC)

$$\begin{cases} \mu(x_1 + x_2 + x_3 - 1) = 0 \\ s_1x_1 = 0 \quad (\text{or } (4x_1 + x_2 - \lambda - \mu)x_1 = 0) \\ s_2x_2 = 0 \quad (\text{or } (8x_2 + x_1 + 2x_3 - \lambda - \mu)x_2 = 0) \end{cases}$$